Assignment - 2

Q Fill in the blanks:-

1 The Enity-Relation Diagrisually represents how entities relate in a database, offering a data blueprint of the data's organizational structure

2. In the Relational model, data is organized into tables with relationship b/w entities defined through foreign key constraints.

3. The Object-Oriented Model treats as a collection of object, each with unique attributes and material/methods, mirroring real-words entities and interactions

4. Integrity constraints ensure the realiability of data by enforcing rules that must be adhered to, maintaining the accuracy of information

Q2 In a Healthcare system, discuss how data structure/abstraction can simplify the interaction between healthcare professional and patient records.

Answer: Data abstraction helps simplify complex healthcare database by hiding unnecessary details and showing only relevant information to users. Doctors, nurses and administrative staff interact with patient record through simplified interfaces without needing to understand how the data is stored internally.

for example, a doctor can view patient history, prescriptions and test results through a medical system interface without knowing how the database tables or relationship are structured. This improve efficiency and reduces the chance of error. Data abstraction also enhanced

security by restricting access to sensitive information base on user roles. Thus, healthcare professional can focus on patient care rather than database complexity.

Q3 Explore the potential benefits and challenges associated with implementing data abstraction in this context.

Answer Benefits

→ Simplified Data Access: Healthcare professional can easily access patient information without understanding database structure.

→ Improved Security: Sensitive data can be protected by controlling access at different healthcare data

→ Better Data Management: Data abstraction helps maintain organised and structured healthcare data

→ Reduce Complexity: User interact with a simplified interaction rather than complex database systems.

Challenges

→ Implementation Complexity: Designing abstraction layers can require significant development effort.

→ Performance Overhead: Additional abstraction layers may slightly affect system performance

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→ Maintenance Difficulty: Changes in database structure may require updates in abstraction layers.

→ Training Requirements: Healthcare staff may need training to use new systems effectively.

Q4 Design a database for a university using a Entity-Relationship Model.

Ans: A university Database can include three main entities: Students, Courses, and Professors.

Entities and Attributes

1. Student
   - Student_ID (Primary key)
   - Name
   - Email
   - Departement
   - Year.
2. Courses
   - Course_Name
   - Course_ID (Primary key)
   - Credits
   - Departement
3. Professor
   - Professor_ID (Primary key)
   - Name
   - Email.
   - Departement.

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Relationships

Enrollement (Student - Course)

A student can enroll in multiple courses. A courses can have many students. This is a many to many Relationship.

Teaching (Profesor - courses).

A professor can teach multiple courses. Each courses is taught by one professor. This is a one to many relationship.

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